Sri Lanka Institute of Information Technology (SLIIT)
Faculty of Computing

SE2030 – Software Engineering
Lab Sheet 04 – Activity Diagrams
Year 2 Semester 1 (2026)

Student ID	Name
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IT21158872	Vimukthi W.M.B.V

Group ID : Group 2
Submission Date : 30th March 2026
Web-Based Library Management System

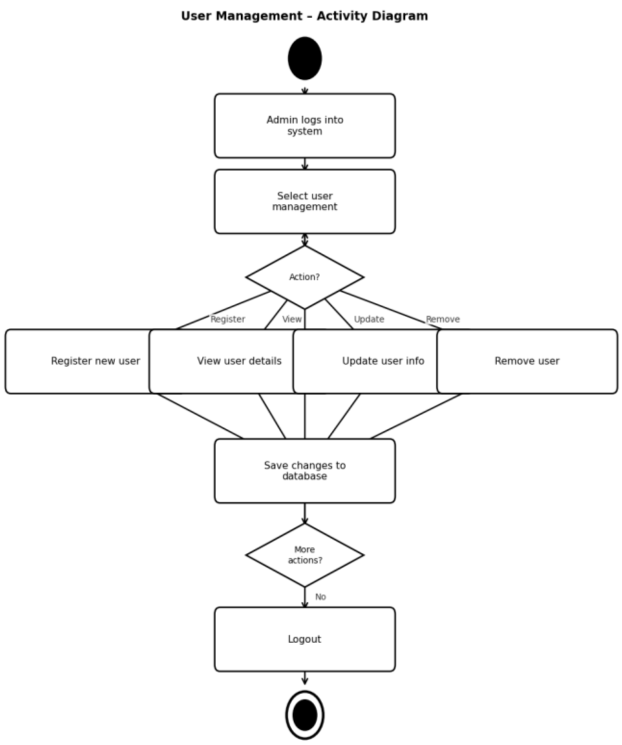
1. User Management

Assigned Member : IT21077524 – Pathirana A.P.C.E

1.1 Use Case Scenario

Use Case Name	Manage User Accounts
Actor	Administrator / Senior Librarian
Precondition	Admin is logged into the system
Main Flow	1. Admin selects the User Management module. 2. System displays the list of existing users. 3. Admin chooses to register, view, update or remove a user. 4. If registering – Admin fills in user details and submits. 5. System validates and saves the new user to the database. 6. If updating – Admin selects a user and modifies details. 7. If removing – Admin selects a user and confirms deletion. 8. System updates the database and confirms the action.
Alternate Flow	If required fields are empty, system shows a validation error and prompts the admin to correct the input.
Postcondition	User database is updated with the latest changes.

1.2 Activity Diagram



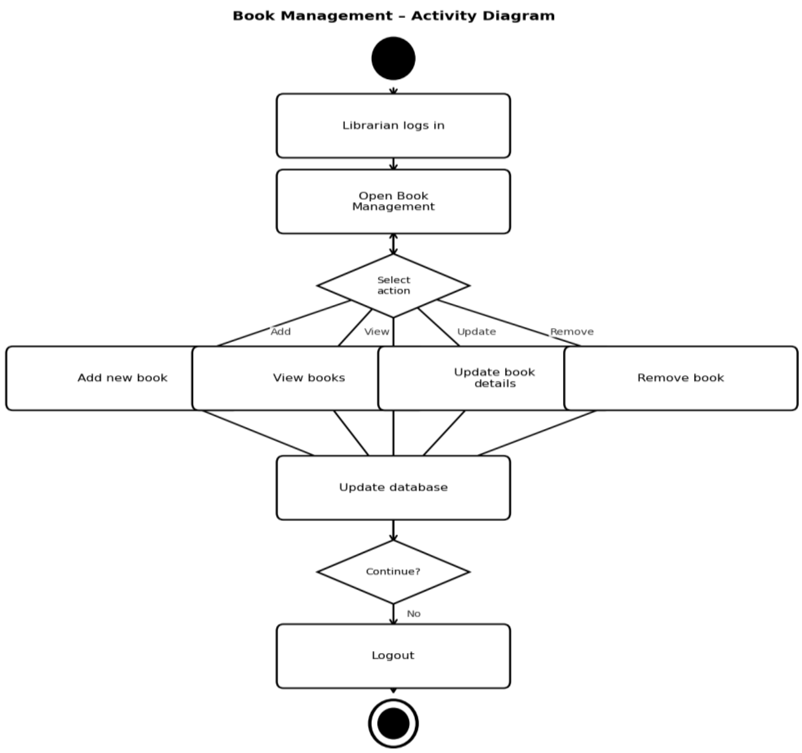
2. Book Management

Assigned Member : IT21054372 – Samarasinghe P.D.P

2.1 Use Case Scenario

Use Case Name	Manage Book Records
Actor	Senior Librarian / Assistant Librarian
Precondition	Librarian is logged into the system
Main Flow	1. Librarian opens the Book Management module. 2. System displays current book records. 3. Librarian selects an action – Add, View, Update or Remove. 4. If adding – Librarian enters title, author, ISBN and copies. 5. System validates and saves the book to the database. 6. If updating – Librarian selects a book and edits the details. 7. If removing – Librarian confirms removal of the selected book. 8. System updates and confirms changes in the database.
Alternate Flow	If ISBN already exists, system shows a duplicate error and cancels the addition.
Postcondition	Book database is accurate and up to date.

2.2 Activity Diagram



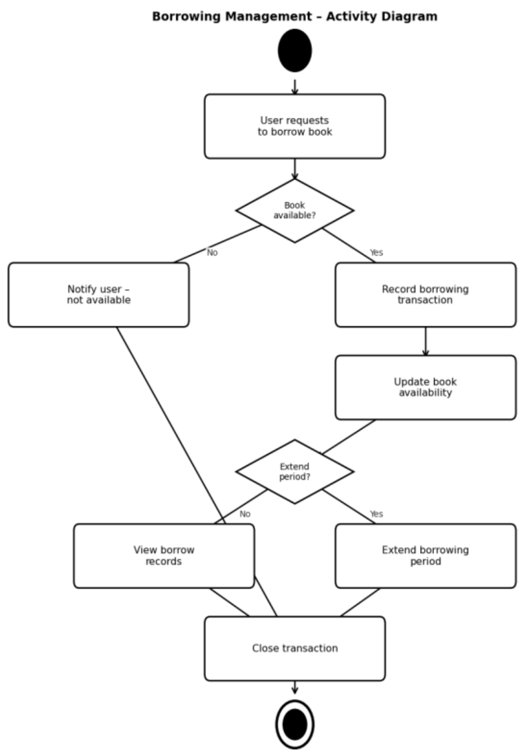
3. Borrowing Management

Assigned Member : IT21158872 – Vimukthi W.M.B.V

3.1 Use Case Scenario

Use Case Name	Borrow a Book
Actor	Undergraduate Student / Librarian
Precondition	User is logged in; book is available in the library
Main Flow	1. User requests to borrow a specific book. 2. System checks book availability. 3. If available – System records the borrowing transaction. 4. System sets the due date for return. 5. Book availability is updated in the database. 6. Librarian can view all active borrowing records. 7. If needed, borrowing period can be extended.
Alternate Flow	If book is not available, system notifies the user and suggests making a reservation.
Postcondition	Borrowing transaction is recorded and book availability is updated.

3.2 Activity Diagram



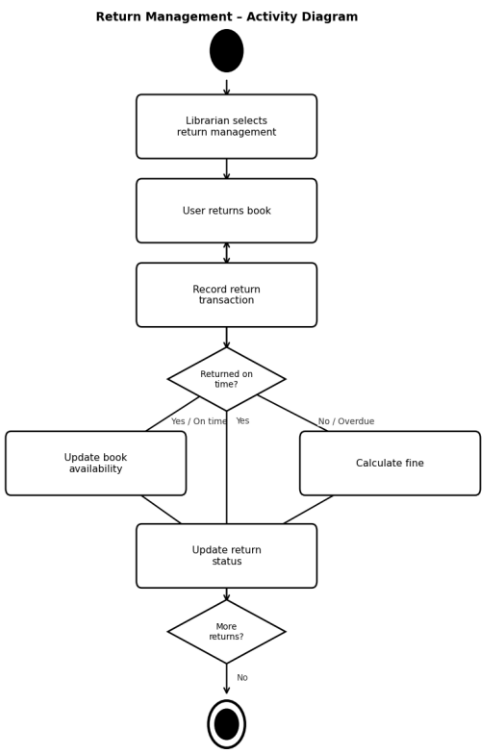
4. Return Management

Assigned Member : IT21047756 – Weerasekara W.B.M.K.D

4.1 Use Case Scenario

Use Case Name	Return a Borrowed Book
Actor	Librarian / Library Staff
Precondition	Book has been borrowed and user is returning it
Main Flow	1. Librarian opens the Return Management module. 2. Librarian selects the relevant borrowing record. 3. System records the return transaction with return date. 4. System checks whether return is on time or overdue. 5. If overdue – fine is calculated automatically. 6. Book availability is updated in the database. 7. Return status is marked as completed.
Alternate Flow	If return record is entered incorrectly, librarian can update or remove the incorrect entry.
Postcondition	Book is marked as available and return records are updated.

4.2 Activity Diagram



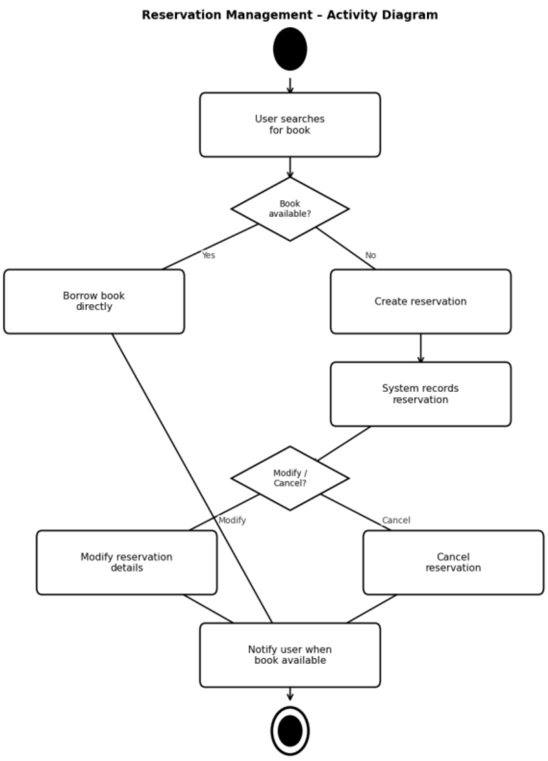
5. Reservation Management

Assigned Member : IT21060144 – Rathnayake R.M.I

5.1 Use Case Scenario

Use Case Name	Reserve a Book
Actor	Undergraduate Student / Lecturer
Precondition	User is logged in; required book is currently unavailable
Main Flow	1. User searches for a book. 2. System shows book as currently unavailable. 3. User selects to create a reservation. 4. System records the reservation details. 5. User can view or modify reservation details. 6. When book becomes available, system notifies the user. 7. User can cancel reservation if no longer needed.
Alternate Flow	If book is available, user is redirected to borrow it directly instead of reserving.
Postcondition	Reservation is saved and user is notified when book is available.

5.2 Activity Diagram



6. Fine Management

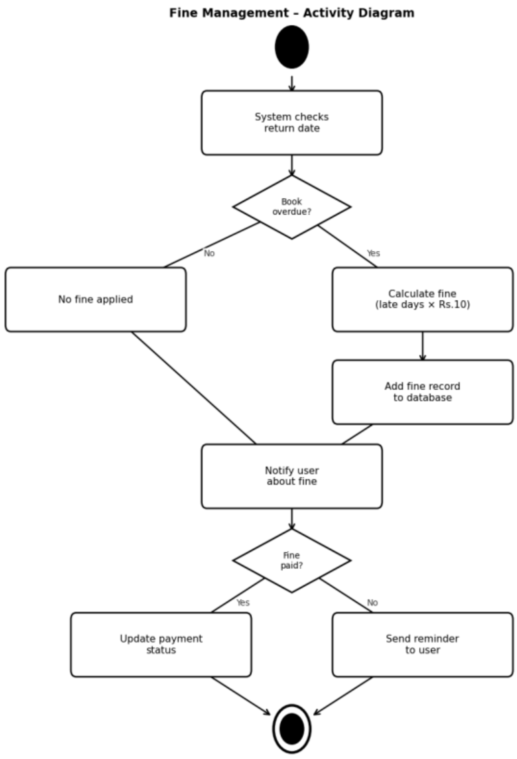
Assigned Member : IT21101274 – Dolawatta T.Y.R

6.1 Use Case Scenario

Use Case Name	Manage Overdue Fines
Actor	Librarian / Student Services Officer
Precondition	Book return date has passed without return
Main Flow	1. System automatically checks return dates daily. 2. If a book is overdue, fine is calculated. 3. Fine = Late Days × Rs.10 per day. 4. System adds fine record to database. 5. User is notified about the outstanding fine. 6. Librarian can view all fine records. 7. When user pays, librarian updates the payment status.
Alternate Flow	If a fine was entered incorrectly, librarian can remove or update the fine record.
Postcondition	Fine records are accurate and payment status is up to date.

Fine Formula : Fine = Late Days × Fine per Day (Rs.10)
Example : Due Date = 14 June, Return Date = 16 June → Late Days = 2 → Fine = Rs.20

6.2 Activity Diagram


Fine Management – Activity Diagram
System checks return date
Book overdue?
No
Yes
No fine applied
Calculate fine (late days × Rs.10)
Add fine record to database
Notify user about fine
Fine paid?
Yes
No
Update payment status
Send reminder to user